Product Development Lifecycle

Capulus Technologies - Product Development Lifecycle
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1.0 Overview
This document outlines the stages, deliverables, and quality gates for developing AI■powered products at Capulus, from ideation to retirement.

2.0 Lifecycle Phases
2.1 Discovery and Ideation
• Activities: Market research, customer interviews (law enforcement, city planners), feasibility study.
• Deliverables: Product concept document, initial business case.
• Gate: Concept approval by Product Council.

2.2 Requirements and Design
• Activities: Define user stories, success metrics, data requirements, and ethical considerations.
• Deliverables: PRD (Product Requirements Document), UX prototypes, data sourcing plan.
• Gate: Design review with Engineering, Security, and AI Ethics Board.

2.3 Development and Testing
• Activities: Agile sprints, continuous integration, unit testing, model training/evaluation.
• Deliverables: Working software, test reports, bias assessment results.
• Gate: Code freeze and pre■launch security review.

2.4 Pilot / Limited Deployment
• Activities: Deploy to a small set of real users (e.g., one police district), collect feedback, monitor performance.
• Deliverables: Pilot evaluation report, incident log.
• Gate: Pilot success review; decision to scale.

2.5 General Availability (GA)
• Activities: Full deployment, marketing launch, user training.
• Deliverables: GA release notes, monitoring dashboards.
• Gate: Post■launch review at 30/60/90 days.

2.6 Maintenance and Retirement
• Activities: Ongoing support, patches, feature updates; end■of■life planning.
• Deliverables: Deprecation notice, data migration plan.

3.0 AI■Specific Requirements
• Data Governance: All training data must be documented and approved for use.
• Model Cards: Each AI model must have a model card detailing its intended use, performance metrics, and limitations.
• Human Oversight: For high■risk applications (e.g., facial recognition in policing), a human must always confirm AI recommendations.

4.0 Quality Gates and Approvals
Phase | Required Approvals

| ------- | ------------------- |
Discovery | Product Manager + VP Product

Design | Engineering Lead, Security, AI Ethics

Development | QA Lead, Product Owner

Pilot | Customer reference, Legal

GA | Executive Committee

5.0 Metrics for Success
• Customer adoption rate.
• Model accuracy / precision / recall.
• Time from concept to GA.
• Post■release defect density.